Prativimb architecture

Hybrid patient twin: mechanistic physiology, learned residual, calibrated uncertainty, plain language reasons.

SOURCES

EHR (FHIR R4 bundles)

Problems, ICD-10

Labs and vitals, LOINC

Medications, ATC

Genetic markers, consent

Wearable stream

CGM every 5 min

Heart rate and steps, 1 min

Sleep stages, nightly RMSSD

Apple Health, Google Fit, Health Connect

Patient reported

Meal log with carbohydrate

Dose confirmations

Declared routine and fasts

Replies from the patient's phone

Synthetic generator

UVA/Padova physiology

Indian diet, Ramadan, shifts

Restartable: the reference

for checking what ifs

INGEST AND STORAGE

Parsers and adapters

FHIR bundle parser

Apple Health, Google Fit,

Health Connect adapters

Unit checks (mmol/L)

DuckDB

EHR tables and features

5 minute wearable grid

Logs, sleep summaries

Seven day windows

Replay service

Demo clock, pause, speed

WebSocket stream

Withdrawn consent stops a source

Logged walks via the reference

Clinical store

Seen, calls, checks, meals, plans,

consent, messages, via FastAPI

Append only audit log

Undo for every change

TWIN

EHR priors

HbA1c, BMI, fasting glucose

TCF7L2, diagnosis, age

give a cold start twin

Twin calibration

Days 1 to 7 of each patient

MAP least squares (scipy)

8 personal parameters

about 1 s per patient

Physics model

Extended Bergman ODE

Meals, exercise, sleep debt,

circadian, shift, metformin

RK4 batch, solve_ivp

LEARNING AND UNCERTAINTY

Cardio ensemble

Windows from DuckDB

Isolation forest, baseline z

LightGBM, stacked by

logistic regression

Conformal layer

Conformalised quantile

regression, 90% band

Alert level chosen on

calibration patients

Residual LightGBM

Learns the twin's error

at 30, 60, 90, 120 min

Features from DuckDB

DECISION SUPPORT

Alert engine

Watch: point above 180

Alert: lower bound above 180

Low glucose

Cardiovascular flag

Explainability

Exact Shapley, twin inputs

TreeSHAP, residual model

mg/dL, then one plain

clinical sentence

What-if simulator

Walk, food, timing, insulin

Tonight, 24 hours, 7 days

Checked against the generator

Plans checked a week later

DELIVERY

FastAPI

REST and WebSocket

Role checks: doctor,

nurse, patient

SOAP note PDF

Dashboard and phone

Ward by urgency, reasons

Today, Try a change, Record

Earlier alerts, outcomes

Handover, daily digest

Festival planner

Patient phone, Hindi,

Tamil, English, Listen

Consent on screen

React, Observable Plot

Hybrid core: physics gives plausibility and extrapolation, the residual model gives accuracy, conformal gives honest uncertainty.

Synthetic data only. Decision support, not a diagnosis.